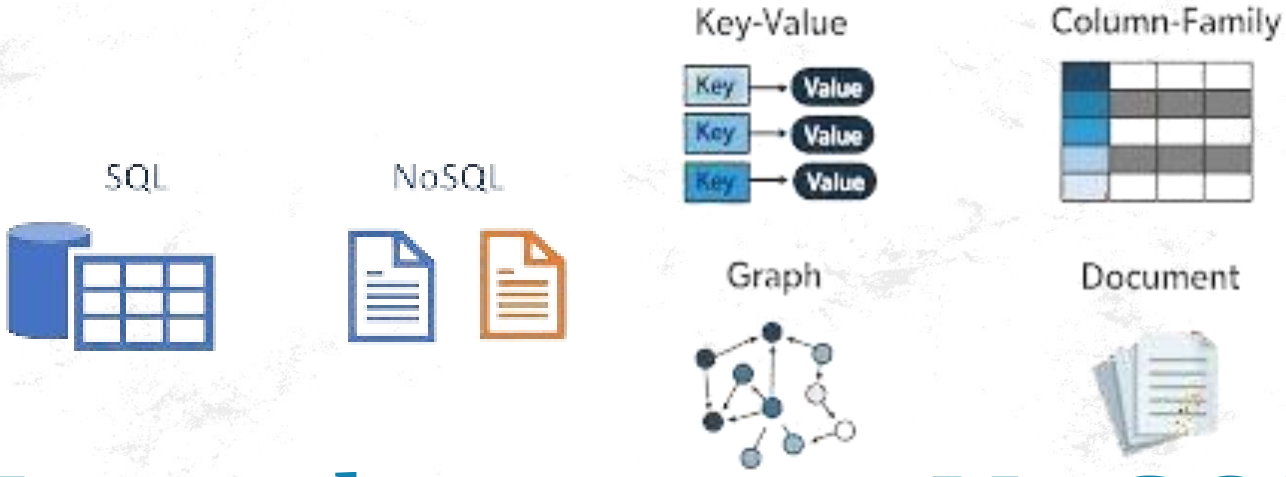


Introduction to **NoSQL**

NIELIT Chandigarh/Ropar

"Users love MongoDB because it offers the fastest time to value compared to any other DBMS technology".

Agenda

- Introduction to NoSQL
- Types of NoSQL Databases
- MongoDB Overview
- MongoDB CRUD Operations
- MongoDB Aggregation Framework
- Practical Hands-on: CRUD & Aggregation in MongoDB
- Creating a Todo (Notes) app using MongoDB

Introduction to NoSQL

What is NoSQL?

- "NoSQL" stands for "Not Only SQL."
- A non-relational database management system designed to handle large amounts of data and diverse data models.

Why NoSQL?

- Scalability
- Flexibility in data modeling
- High performance for specific use cases
- Designed for distributed systems

Key Characteristics:

- Schema-less or flexible schema
- Horizontal scaling
- Optimized for modern data requirements (e.g., big data, real-time analytics)

Types of NoSQL Databases

1. Document Databases

1. Example: MongoDB
2. Data is stored as JSON-like documents.
3. Best for semi-structured data.

2. Key-Value Stores

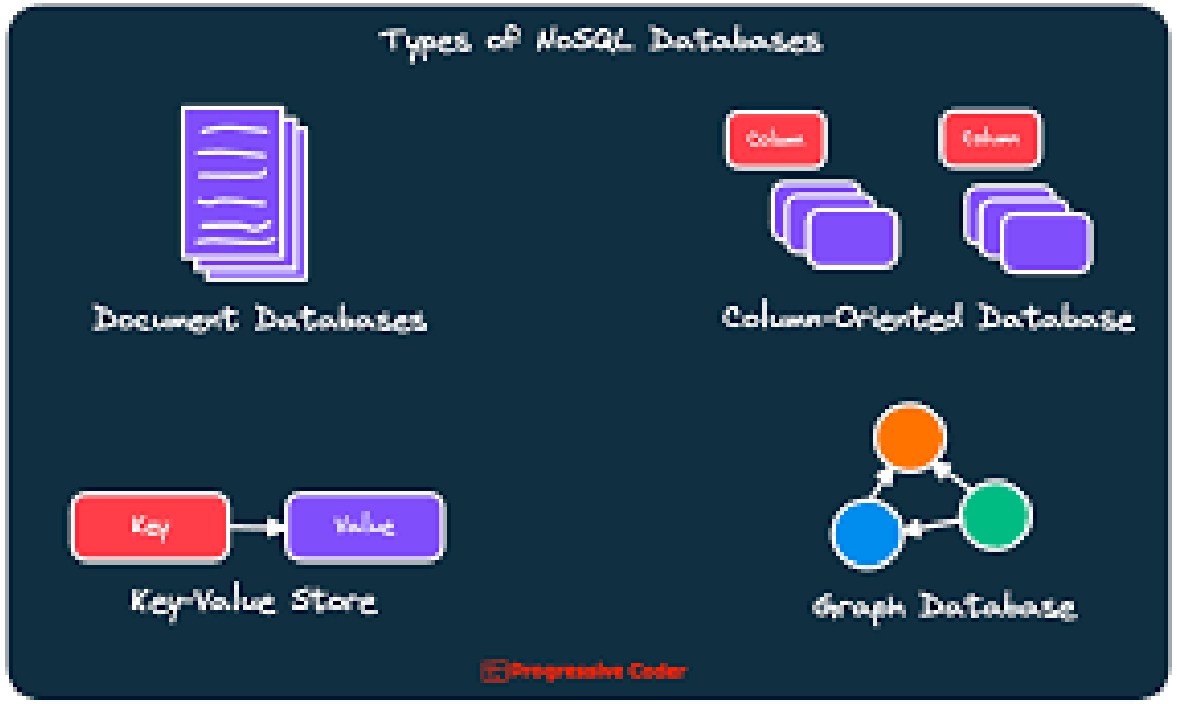
1. Example: Redis, DynamoDB
2. Data is stored as key-value pairs.
3. Ideal for caching and session storage.

3. Column-Oriented Databases

1. Example: Cassandra, HBase
2. Data is stored in columns instead of rows.
3. Suitable for analytical workloads.

4. Graph Databases

1. Example: Neo4j, ArangoDB
2. Data is represented as nodes and edges.
3. Best for relationship-based queries (e.g., social networks).



MongoDB Overview

MongoDB is a popular, open-source, document-oriented NoSQL database.

- Stores data in **JSON-like documents** (BSON).
- Supports **schema-less** data models.
- Provides **horizontal scaling** and high availability through sharding.
- Used by organizations such as Uber, eBay, and Netflix for large-scale applications.

Key Features:

- **Flexible Schema**: Allows for a dynamic and flexible data model.
- **High Availability**: Replica sets for data redundancy and fault tolerance.
- **Sharding**: Distributes data across multiple machines for horizontal scaling.
- **Aggregation Framework**: Powerful data transformation and analysis tool.

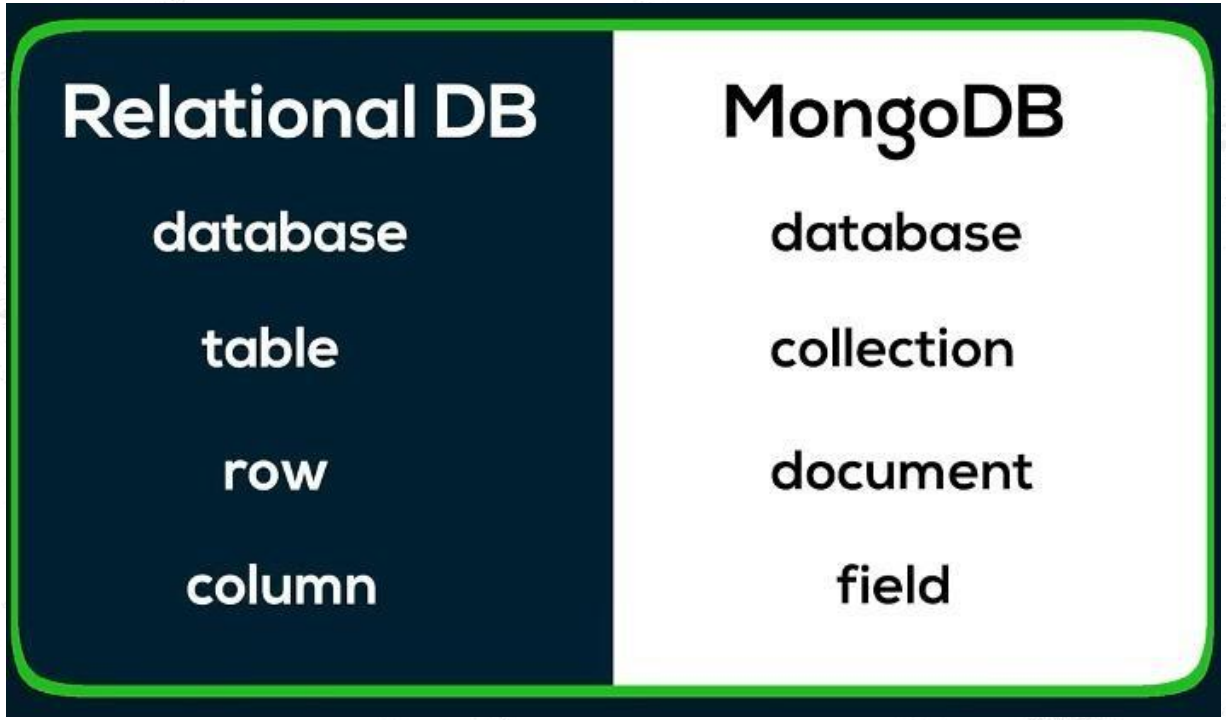
MongoDB Overview

Features:

- Schema flexibility
- Horizontal scaling with sharding
- Built-in replication for high availability
- Rich query language

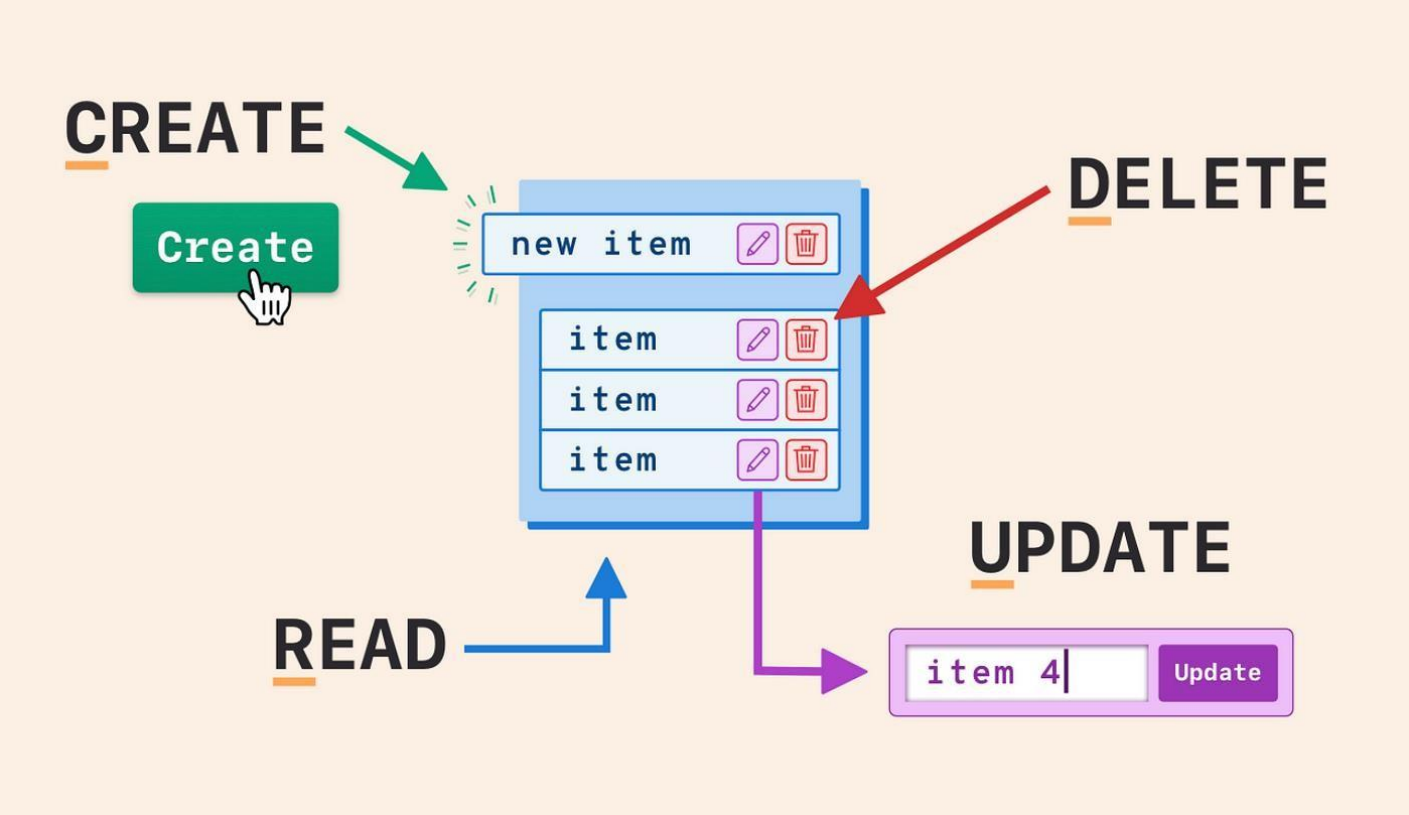
Use Cases:

- Content management
- Real-time analytics
- IoT applications
- Mobile and web applications



MongoDB CRUD Operations - Overview

- **CRUD** stands for Create, Read, Update, and Delete, the basic operations in MongoDB.



Create Operation (Insert)

- MongoDB provides methods to insert documents into collections.
  - **insertOne()**: Inserts a single document.
  - **insertMany()**: Inserts multiple documents.

1. Create:

```
db.collection.insertOne({ key: "value" });

db.collection.insertMany([{ key: "value1" }, { key: "value2" }]);
```

Example:

```
// Insert one document

db.collection.insertOne({

name: "John Doe",

age: 30,

email: "johndoe@example.com" });
```

```
// Insert documents

db.collection.insertMany([

{ name: "Alice", age: 25, email:
"alice@example.com" },

{ name: "Bob", age: 35, email:
"bob@example.com" }]);
```

Read Operation (Find)

- MongoDB provides the **find()** method to query the database.

```
db.users.find({ key: "value" });
db.users.findOne({ key: "value" });
```

Example:

```
// Find all documents in the collection
db.users.find({});
 // Find specific document(s) based on a condition  GT – GREATER THAN
db.users.find({ age: { $gt: 30 } });
```

Projection: Specify fields to return:

```
db.users.find({}, { name: 1, email: 1 });
```

Update Operation (Update)

- MongoDB provides methods to modify existing documents.
  - **updateOne()**: Updates a single document.
  - **updateMany()**: Updates multiple documents.

```
db.users.updateOne({ key: "value" }, { $set: { key: "newValue" } });
db.users.updateMany({ key: "value" }, { $set: { key: "newValue" } });
```

Example:

```
// Update a single document
db.users.updateOne(
{ name: "John Doe" },
{ $set: { age: 31 } } );
```

```
// Update many documents
db.users.updateMany(
{ age: { $lt: 30 } },
{ $set: { status: "young" } } );
```

Delete Operation (Remove)

- MongoDB allows deletion of documents using the deleteOne() and deleteMany() methods.

```
db.users.deleteOne({ key: "value" });
db.users.deleteMany({ key: "value" });
```

Example:

```
// Delete a single document
db.users.deleteOne({ name: "John Doe" });
// Delete multiple documents
db.users.deleteMany({ age: { $lt: 30 } });
```

MongoDB Aggregation Framework

The **Aggregation Framework** is used for data transformation and analysis.

- Supports operations like filtering, grouping, sorting, and reshaping data.

Stages in the aggregation pipeline:

- **$match**: Filters the data.
- **$group**: Groups data by a specific field.
- **$sort**: Sorts data.
- **$project**: Shapes the output (selects specific fields).
- **$limit**: Limits the number of documents.

Aggregation Example - Grouping and Summarizing Data

Example: Grouping users by age and calculating the average age.

```
db.users.aggregate([
 { $group: { _id: "$age", avgAge: { $avg: "$age" } } }
]);
```

Explanation:

- $group: Groups documents by the field age and calculates the average age.

Aggregation Example - Sorting and Limiting

Example: Sorting users by age and limiting the results to the top 5.

```
db.users.aggregate([
 { $sort: { age: -1 } },
 { $limit: 5 }
]);
```

Explanation:

- $sort: Sorts the users by the age field in descending order.
- $limit: Limits the result to the top 5 documents.

MongoDB Aggregation Framework

What is Aggregation?

- A powerful way to perform data processing and analysis.

Stages:

- $match - Filters documents.
- $group - Groups documents by a specified key.
- $project - Shapes the output.
- $sort - Sorts documents.
- $limit - Limits the number of documents.

```
db.collection.aggregate([
{ $match: { status: "active" } },
{ $group: { _id: "$category", total: { $sum: 1 } } },
{ $sort: { total: -1 } }
]);
```

Benefits and Challenges of NoSQL

Benefits:

- High scalability
- Flexible data models
- Fast for specific use cases
- Optimized for big data

Challenges:

- Limited Support for Small Data
- Limited support for complex queries (compared to SQL), Lack of Join-Like Functionality
- Steeper Learning Curve for SQL Users
- Possible consistency trade-offs (e.g., CAP theorem)
- Backup and Restore Complexicity

MongoDB Use Cases

- **Content Management Systems**: Storing and managing unstructured content.
- **E-commerce Platforms**: Product catalogs, user data, shopping cart.
- **Real-Time Analytics**: Storing logs and event data for real-time processing.
- **Social Networks**: Storing and managing user profiles and relationships.
- Artificial intelligence, Edge computing, Internet of things, Mobile, Payments, Serverless development, and Personalization.
- MongoDB is a good choice for teams that need to develop software applications quickly and scale data.

Here are some companies that use MongoDB

- **Walmart**: Uses MongoDB to manage its product catalog
- **Google**: Uses MongoDB Atlas in combination with Google Cloud to build intelligent applications
- **Microsoft**: Integrates MongoDB Atlas with Microsoft Azure's AI and data analytics tools
- **IBM**: Customers can use MongoDB Community Edition and MongoDB Enterprise Edition as fully managed databases in the IBM Cloud
- **Zomato**: An Indian company that uses MongoDB
- **Tata Digital**: An Indian company that uses MongoDB
- **Canara HSBC Life Insurance**: An Indian company that uses MongoDB
- **Tata AIG**: An Indian company that uses MongoDB
- **Devnagri**: An Indian company that uses MongoDB

Conclusion

- **MongoDB** is a powerful, flexible, and scalable NoSQL database.
- **CRUD operations** provide basic functionality for creating, reading, updating, and deleting data.
- **Aggregation** enables complex data processing and transformation.
- MongoDB is ideal for applications that require high scalability, flexibility, and real-time data analysis.

Next Steps:

- Set up MongoDB locally or **on a cloud service like Atlas**.
- Practice CRUD operations and aggregation queries.

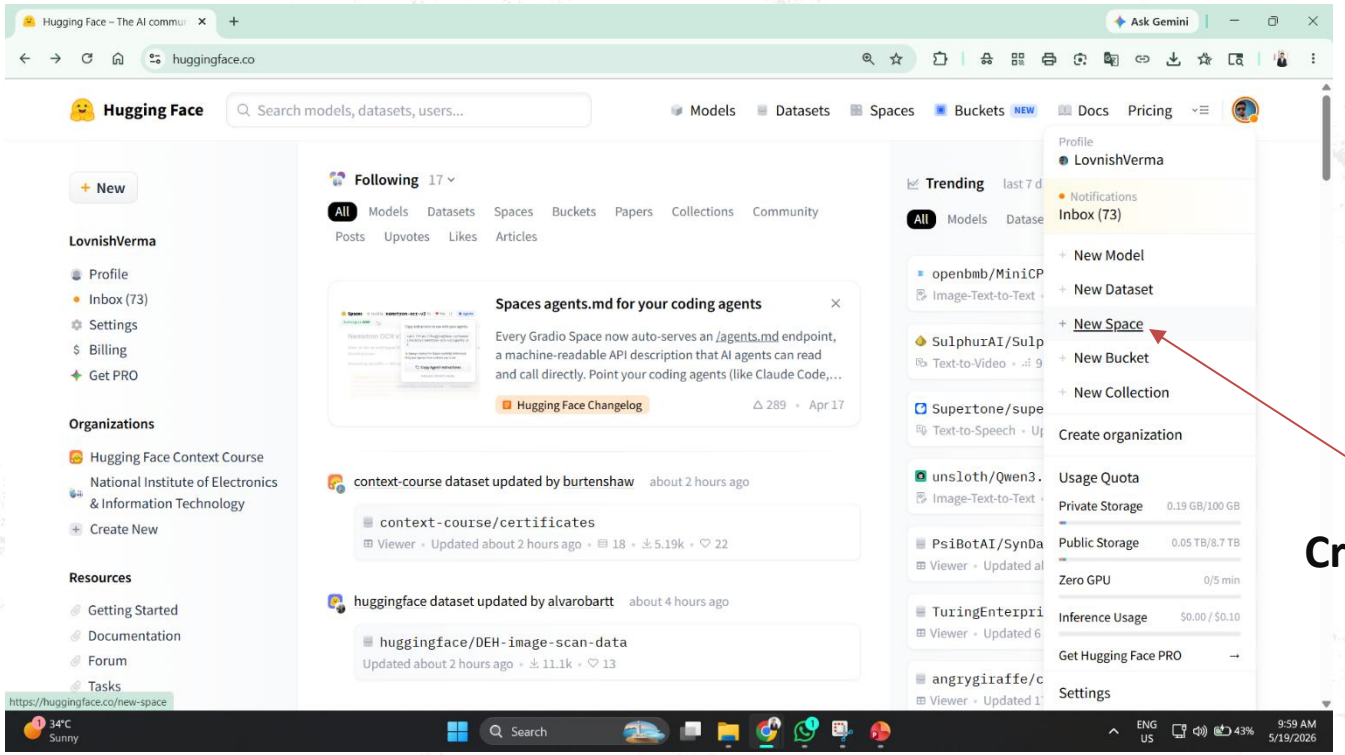
Create Live Project

NIELIT Chandigarh/Ropar

"Users love MongoDB because it offers the fastest time to value compared to any other DBMS technology".

Create a ToDO App using MongoDB and Python Flask Framework On https://huggingface.co/

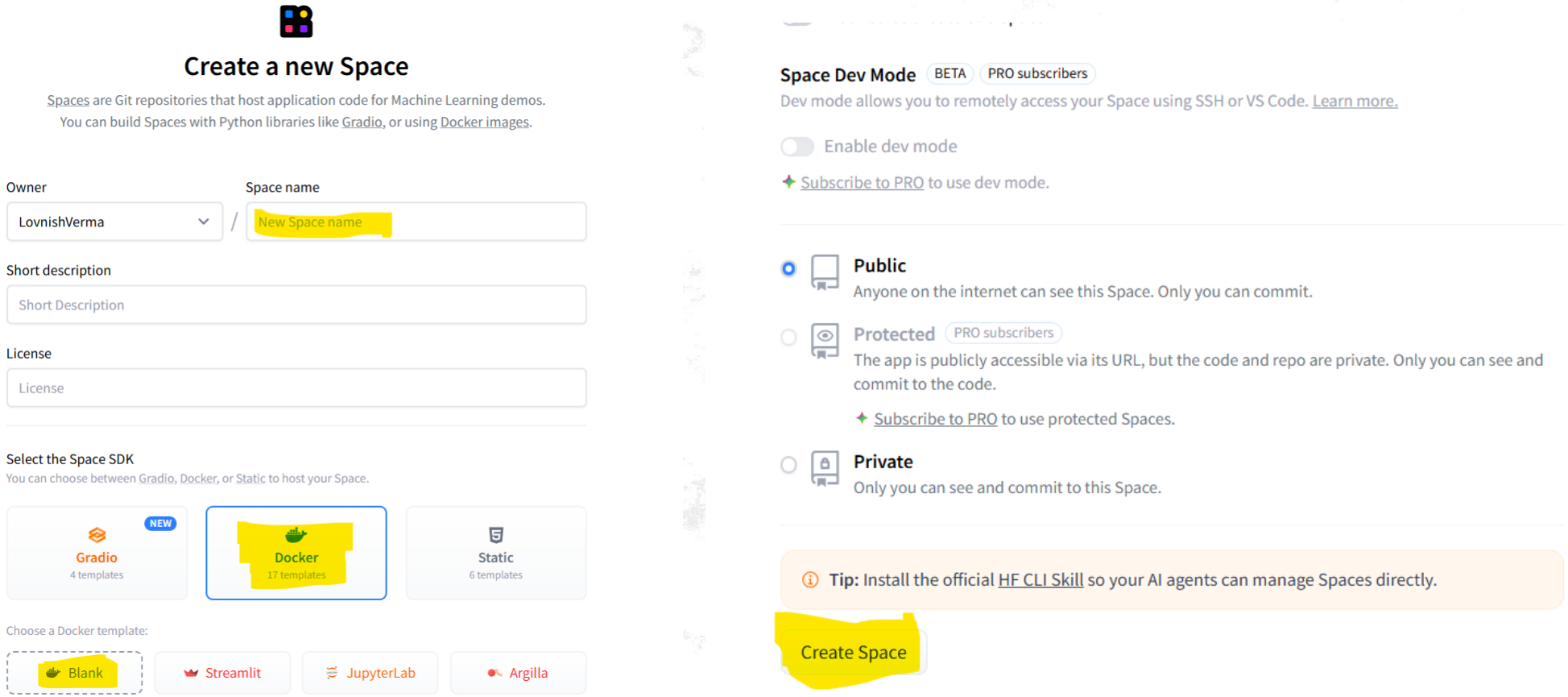
- **Set Up Huggingface Space**:
  - Sign up or Login on huggingface.co
  - Go to huggingface.co homepage after logging in and click on **Create a New Space**.



Create a New Space.

Create a ToDO App using MongoDB and Python Flask Framework On Huggingface.co

- Enter Space Name
- Select Docker as SDK / Keep Template Blank



Create a ToDO App using MongoDB and Python Flask Framework On Huggingface

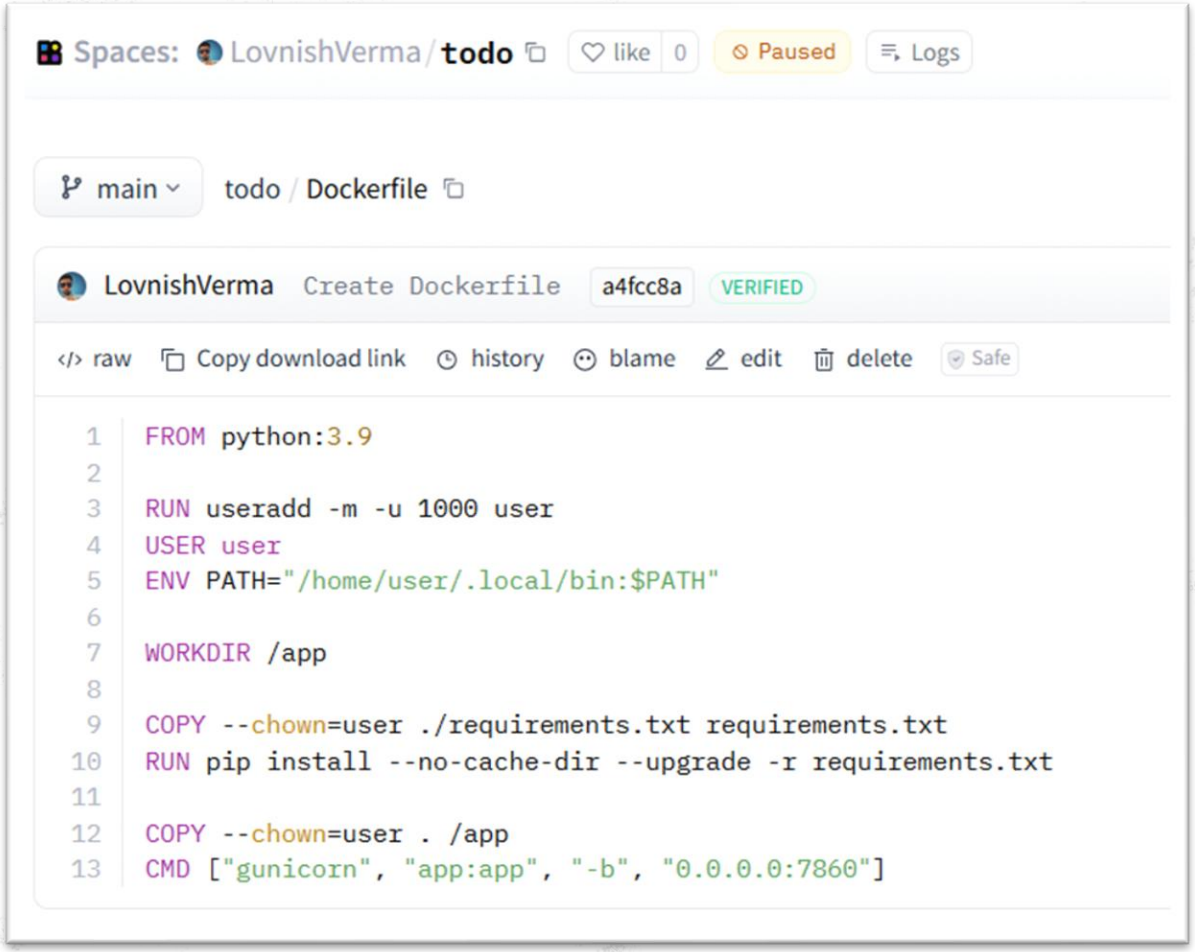
Scroll down a little and Copy all this code

Create your Dockerfile:

```
# Read the doc: https://huggingface.co/docs/hub/spaces-sdks-docker
# you will also find guides on how best to write your Dockerfile

FROM python:3.9

RUN useradd -m -u 1000 user
USER user
ENV PATH="/home/user/.local/bin:$PATH"

WORKDIR /app

COPY --chown=user ./requirements.txt requirements.txt
RUN pip install --no-cache-dir --upgrade -r requirements.txt

COPY --chown=user . /app
CMD ["uvicorn", "app:app", "--host", "0.0.0.0", "--port", "7860"]
```

Hint Alternatively, you can create the Dockerfile file directly in your browser.

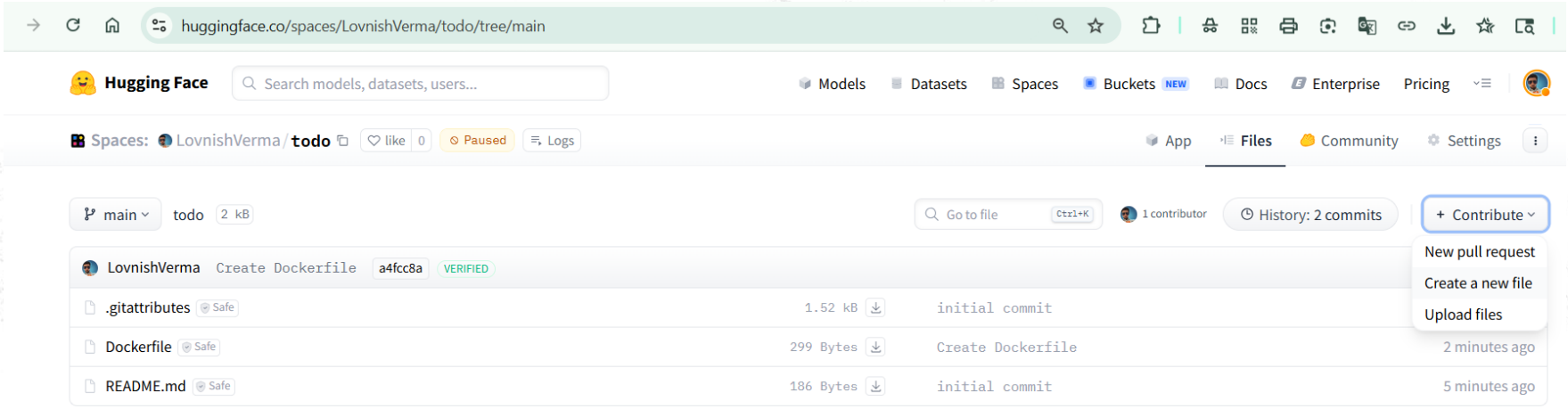
Click on create the Dockerfile



CMD ["gunicorn", "app:app", "-b", "0.0.0.0:7860"]

Create a ToDO App using MongoDB and Python Flask Framework On Huggingface

- Click on **Files** and then **Contribute** and then **Create a new file**



PROJECT STRUCTURE

```
/my-todo-app
├── app.py
├── requirements.txt
├── Dockerfile
└── /templates
    └── index.html
```

Create a ToDO App using MongoDB and Python Flask Framework on Huggingface

app.py This file connects to MongoDB, fetches tasks, and allows you to add or delete them.

todo/ app.py

Edit Preview

```python
from flask import Flask, render_template, request, redirect, url_for
from pymongo import MongoClient
from bson.objectid import ObjectId
import os

app = Flask(__name__)

# Connect to MongoDB using a secure environment variable
# We will set MONGO_URI in Hugging Face settings later
MONGO_URI = os.getenv("MONGO_URI", "mongodb+srv://test:test@cluster0.sxci1.mongodb.net/?retryWrites=true&w=majority")
client = MongoClient(MONGO_URI)

# Select the database and collection
db = client.todo_database
todos_collection = db.todos

@app.route('/')
def index():
    # Fetch all tasks from MongoDB
    tasks = list(todos_collection.find())
    return render_template('index.html', tasks=tasks)

@app.route('/add', methods=['POST'])
def add():
```

Get it for free from mongodb atlas

Create a ToDO App using MongoDB and Python Flask Framework On Huggingface

- Edit index.html file add following code:

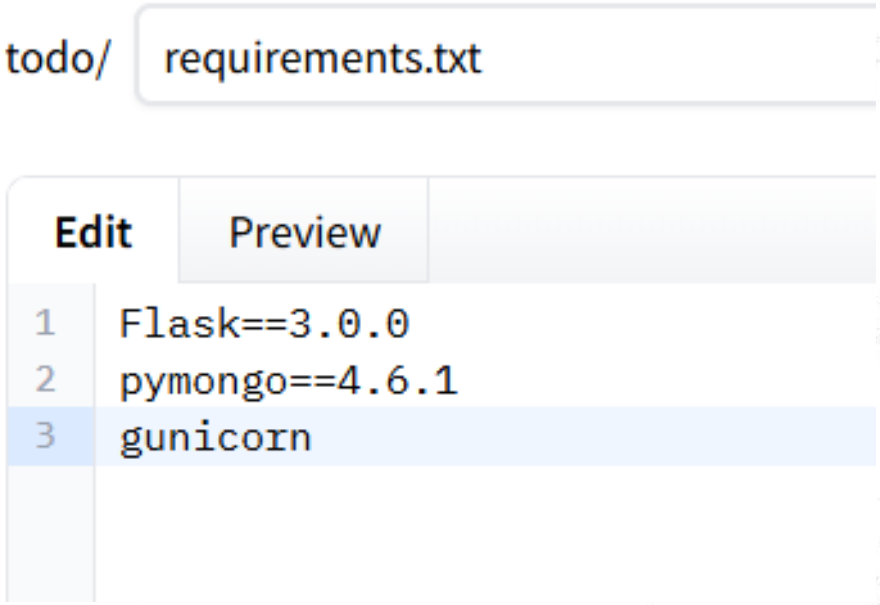
todo/ templates/index.html

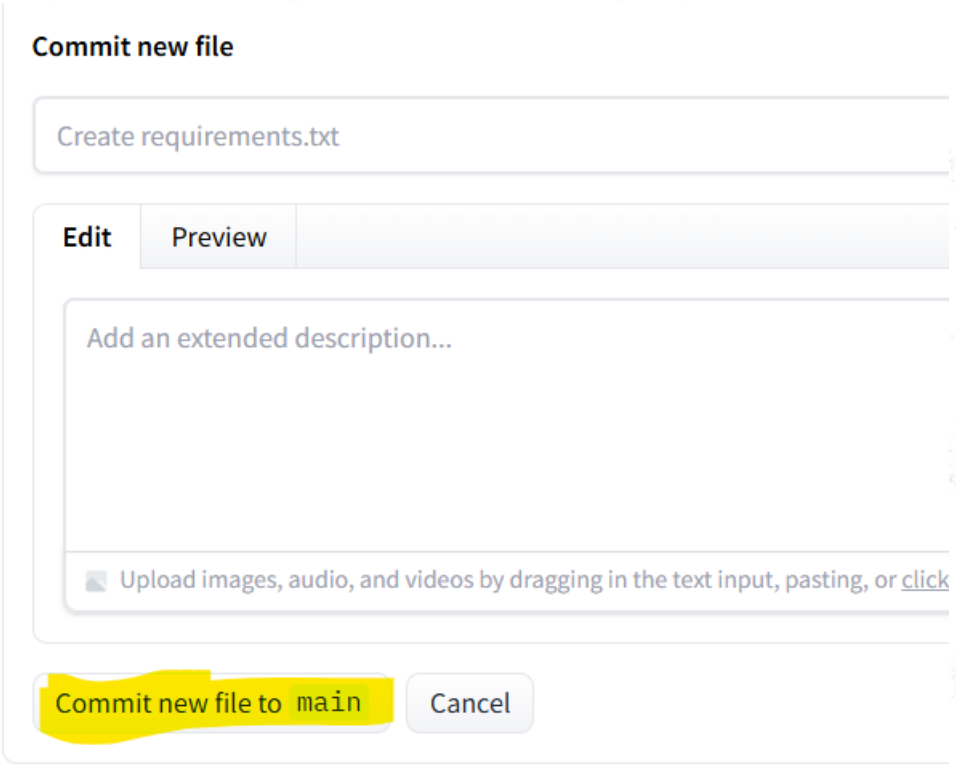
Edit | Preview

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>MongoDB To-Do App</title>
    <style>
        body { font-family: Arial, sans-serif; margin: 40px auto; max-width: 400px; text-align: center; }
        input[type="text"] { padding: 10px; width: 65%; border: 1px solid #ccc; border-radius: 4px; }
        button { padding: 10px 15px; cursor: pointer; background: #198754; color: white; border: none; border-radius: 4px; }
        button:hover { background: #157347; }
        ul { list-style-type: none; padding: 0; }
        li { background: #f8f9fa; margin: 10px 0; padding: 12px; display: flex; justify-content: space-between; border-radiu
        a { color: red; text-decoration: none; font-weight: bold; }
    </style>
</head>
<body>
    <h2>🌿 MongoDB To-Do List</h2>

    <form action="/add" method="POST">
        <input type="text" name="task" placeholder="Enter a new task..." required>
        <button type="submit">Add Task</button>
    </form>
```

Create a ToDO App using MongoDB and Python Flask Framework On Huggingface

- Add list of all required libraries in requirements.txt file.





Steps to get your MongoDB Atlas URL

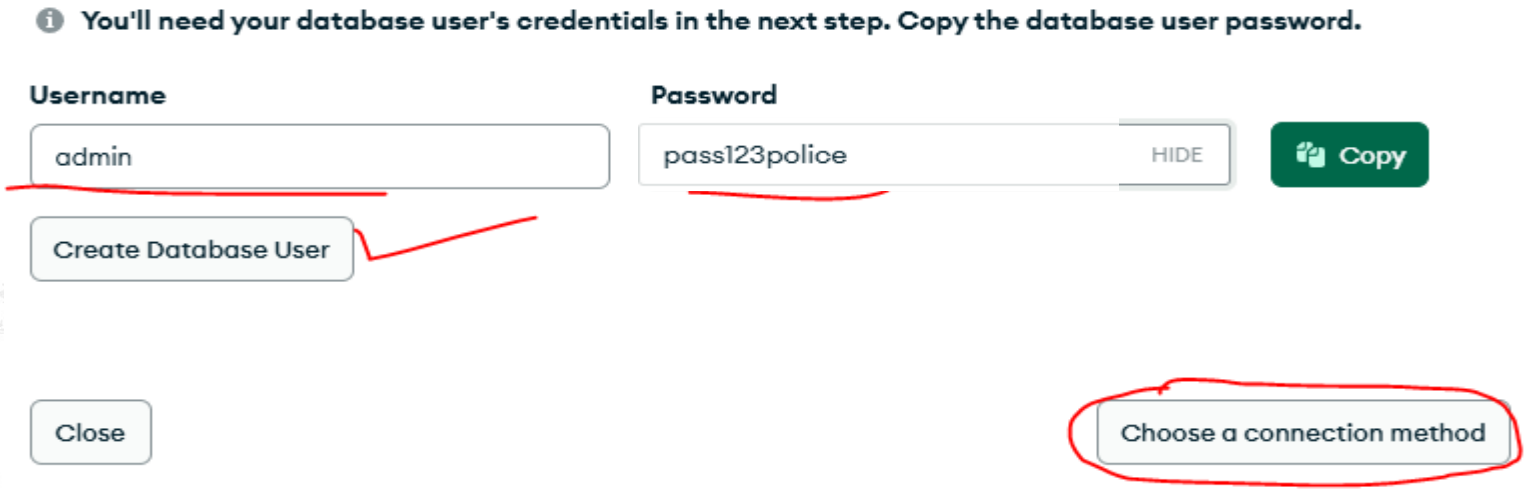
- **1. Sign Up / Log In to MongoDB Atlas**
- Visit MongoDB Atlas.
- If you don't already have an account, sign up for a free account.
- Log in to your account.
- **2. Create a New Cluster**
- Once logged in, click **"Build a Cluster"** or **"Create"**.
- Select a **M0** free tier cluster
  - Set Cluster Name
  - Choose a cloud provider (e.g., AWS, Azure, GCP).
  - Select a region (choose one closer to your location for better performance).
  - Click **"Create Deployment"**.

Steps to get your MongoDB Atlas URL

- **Create a Database User**
- Create a username and password (e.g., admin / pass123police).

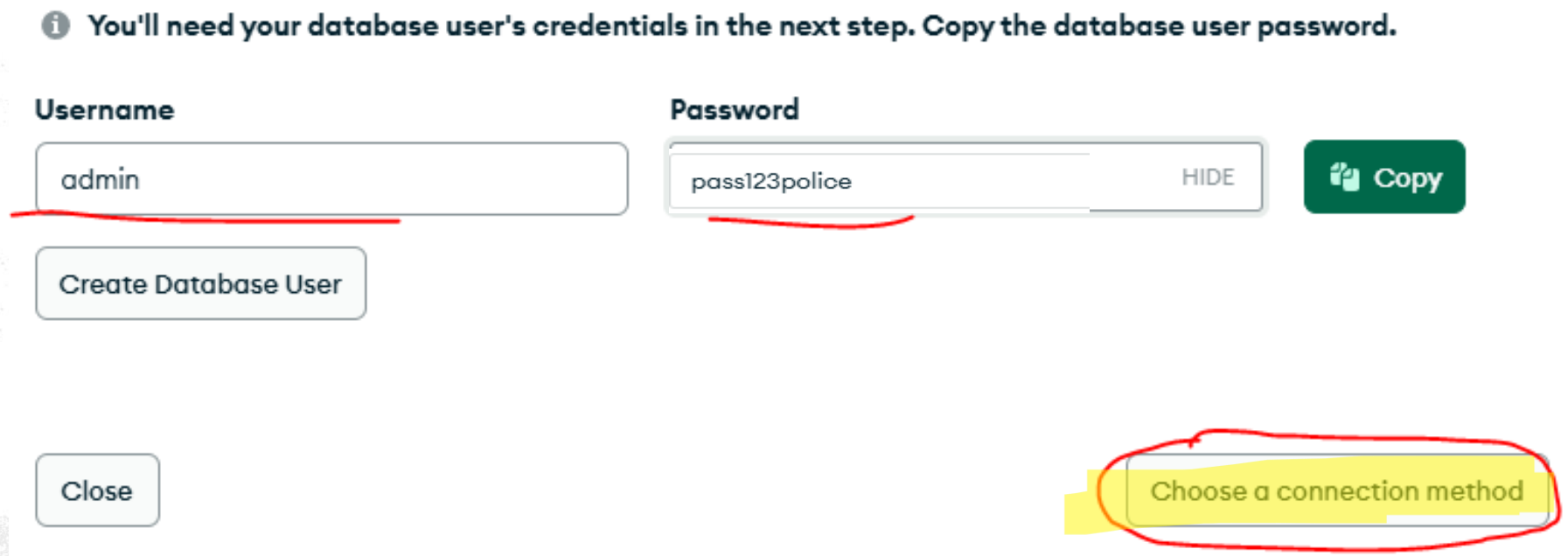
Don’t add @ in your password it will cause error later on.

- Then Click on Create Database User.
- Then click on Choose connection method.



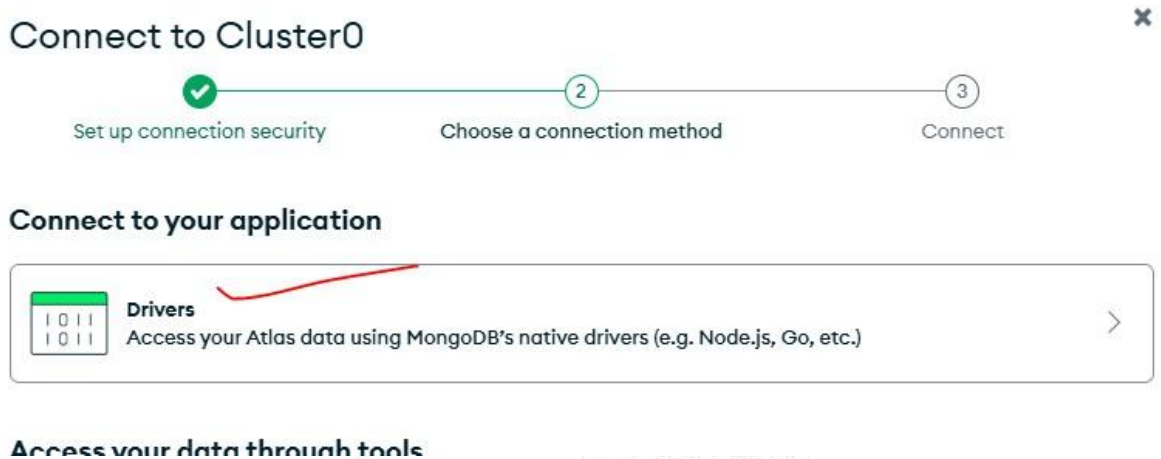
Steps to get your MongoDB Atlas URL

- **Create a Database User**
- Create a username and password (e.g., admin / pass123police).
- Then click on Choose connection method.

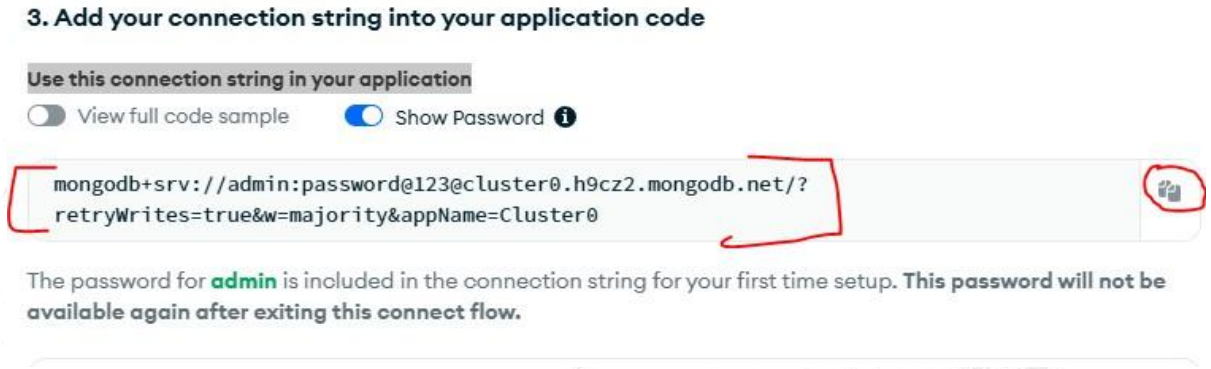


Steps to get your MongoDB Atlas URL

- **Click on Drivers:**

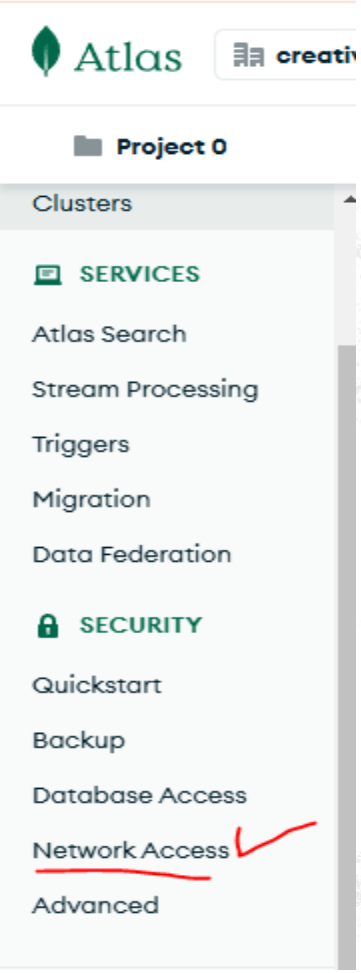


- **You'll See your connection string copy it and save it somewhere safe:**



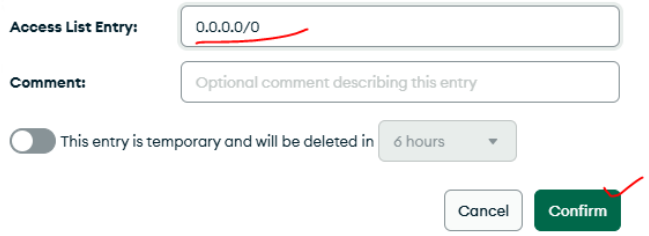
Steps to get your MongoDB Atlas URL

- **Allow Network Access**
- Go to **"Network Access"** in the left sidebar.
- Click **"Add IP Address"**.



  - Choose **"Allow Access from Anywhere"** (recommended for development only). This adds 0.0.0.0/0 to allow access from any IP address.

  - Alternatively, add specific IPs if you know your development machine's public IP.



- Save changes by clicking Confirm.

Create a ToDO App using MongoDB and Python Flask Framework On Huggingface

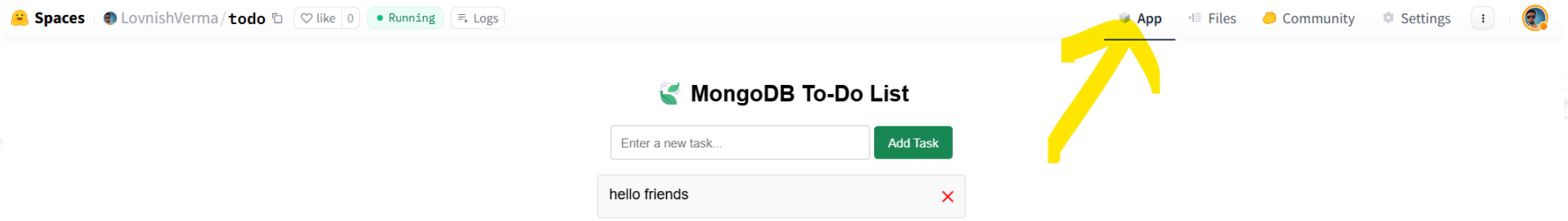
- **Add your Connection URL in your flask app edit app.py file**

```
app = Flask(__name__)
app.secret_key = "nielit_secret_key_here"  # Required for flash messages

# MongoDB connection
MONGO_URI = os.getenv("MONGO_URI", "mongodb+srv://test:test@cluster0.sxci1.mongodb.net/?retryWrites=true&w=majority")
client = MongoClient(MONGO_URI)

# Database and Collection setup
db = client.todo_app  # Database name
todos = db.todos      # Collection name
```

While creating a file write templates/index.html it will create templates folder and inside it index.html file

Create a ToDO App using MongoDB and Python Flask Framework On Huggingface

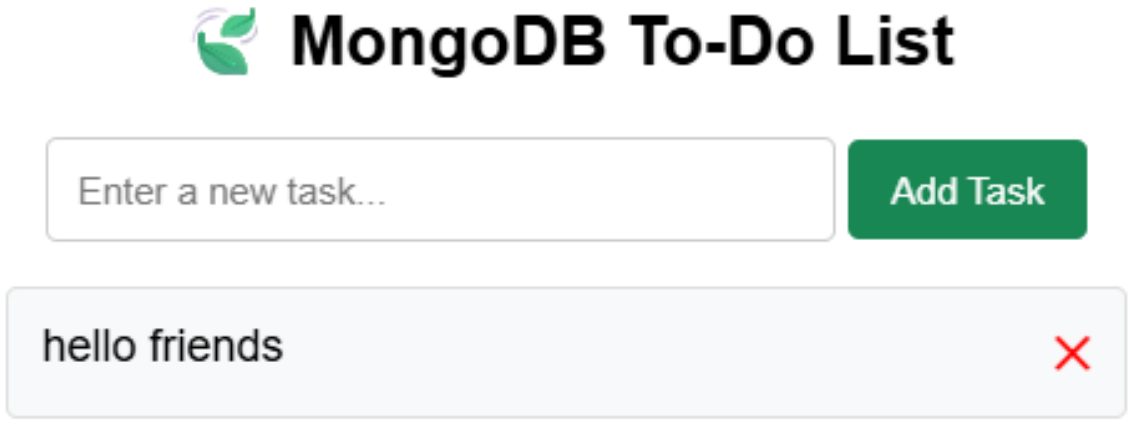
- Let’s View our app Click on app



- You can also see your app by visiting https://yourusername-spacename.hf.space

Create a ToDO App using MongoDB and Python Flask Framework On Huggingface

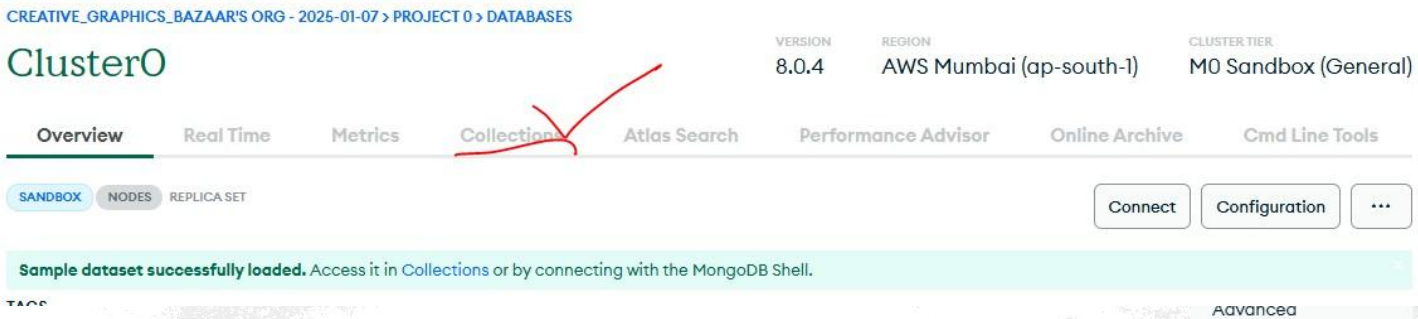
- Let’s Add Some Tasks to test our newly created app.
- Enter Task and Click on Add Task.
- Wow as we can see your app is working Fine.

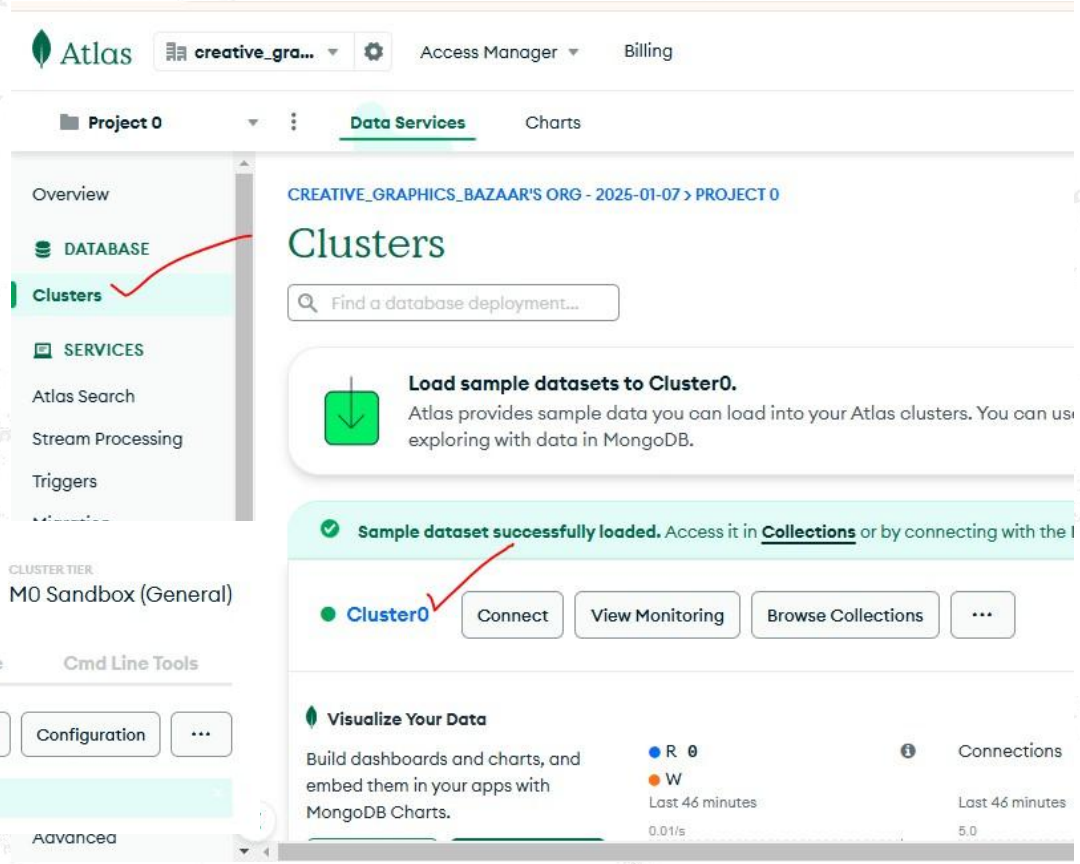


See Collection on MongoDB Atlas

To See your Collection on MongoDB Atlas

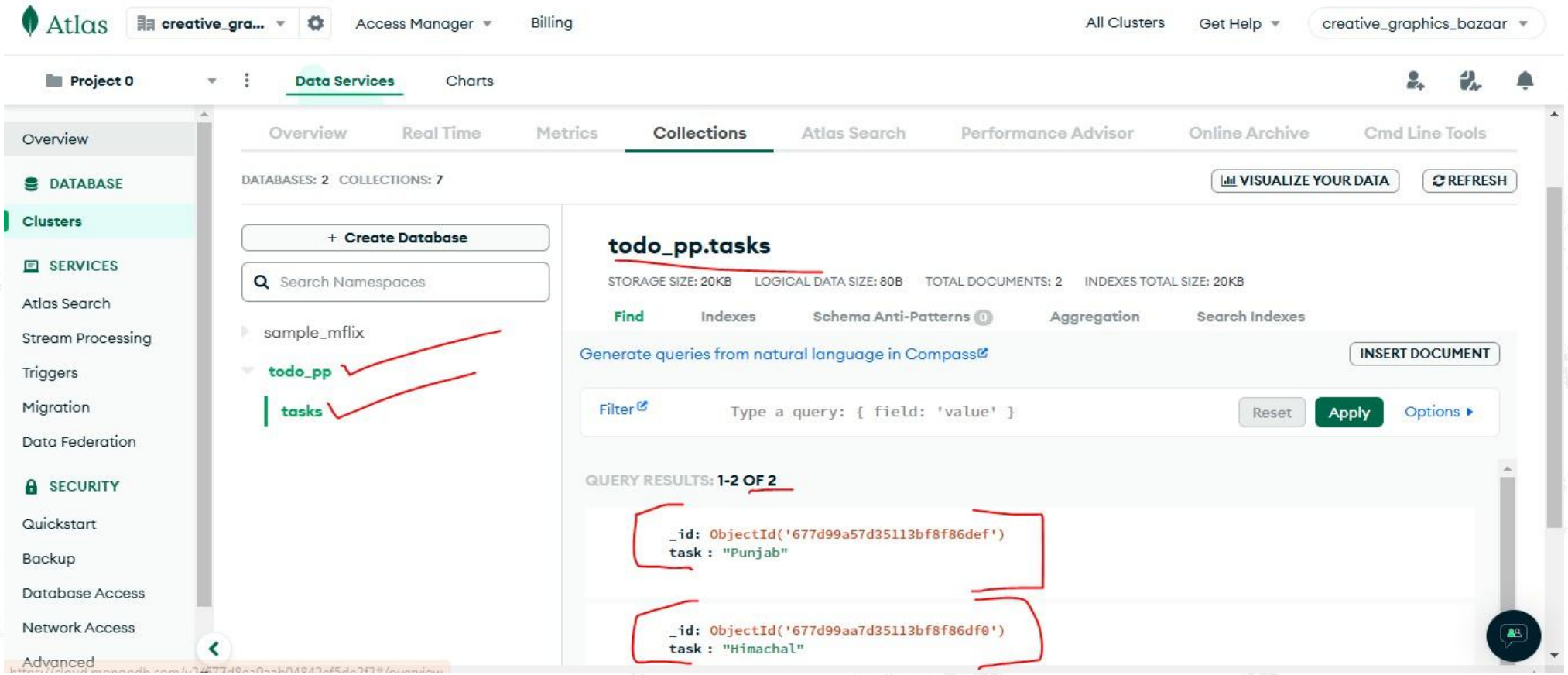
- Click on Clusters.
- Then Click on Your Cluster Name.
- Then Click on Collections.



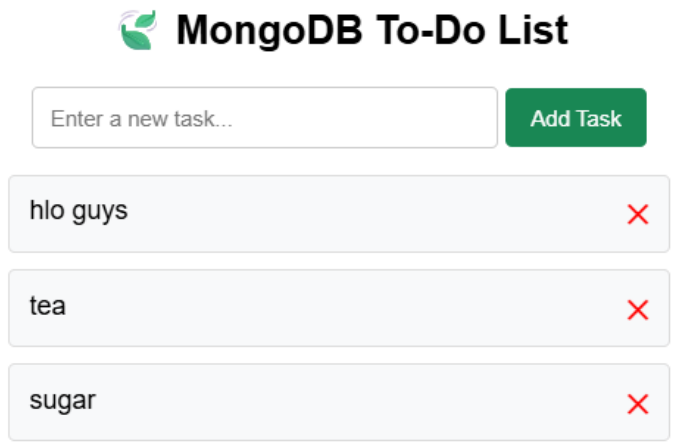


See Collection on MongoDB Atlas

Click on your DB name and then your Collection Name to view all your Collections.



Enjoy your own Personal ToDo App



What's Next:

- Add CSS Styling to improve App appearance.
- Add BootStarp to make it Mobile Fiendly.
- Add Advance Features like aggregation and make your app like https://github.com/lovnishverma/ToDo-List-Flask-PyMongo this improved version of app with better Styling (UI) and MongoDB aggregation.
- Keep Learning Try Creating Similar apps using MongoDB and Flask on Huggingface.

Live Demo

https://huggingface.co/spaces/LovnishVerma/todo